MODULE 09 · DOMAIN 2 - APPLICATIONS AND INTEGRATION

From Business Goal to Requirements

Turning a business need into functional and infrastructure requirements, measurable success criteria, and a trail from each requirement to its proof.

DOMAIN 2 of 8 EXAM WEIGHT 33.1% MODULE 9 of 36

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No architect lays a brick before the brief. **Requirements come before the model, the tool and the framework.**

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A goal is not a requirement

- **The goal:** "Use AI to speed up repair requests." Nothing here can be built or tested.
- **Who uses it,** and who acts on what it produces?
- **What goes in, what comes out,** and what decision does the output support?
- **How good, how fast, how many, how much,** and what happens when it is wrong?
- **What data does it touch,** where may that data go, and which systems does it connect to?

> THE TAKEAWAY
>
> A requirement is something you could fail. If no result could fail it, it is still a goal.

Functional and infrastructure requirements

FUNCTIONAL

What it must do

- Inputs it accepts, outputs it returns
- Decisions and actions it supports
- Rules it must always follow
- What it must never do

INFRASTRUCTURE

How and where it must run

- Latency and throughput, peaks included
- Availability and failure behaviour
- Data sensitivity, residency, retention
- Integrations, platform, cost budget

ON THE EXAM

Infrastructure requirements are the ones teams forget, and they often decide the architecture.

What each requirement decides

IF THE REQUIREMENT SAYS	IT POINTS TOWARDS
A person is waiting and needs the first words fast	Streaming; a model and effort that meet the latency target
Nobody waits; results are needed by morning	The Message Batches API, at half price
Data must be processed in a particular region	A platform and endpoint that can guarantee that region
Answers must point to their source	Citations, grounded on the supplied documents
A rule is exact, like a date or a sum	Code, not Claude
Tens of thousands of requests at peak	Rate-limit planning, queues, retries

THE TIEBREAKER

Read the requirement first. The answer that names a feature usually follows from it.

Success criteria

- **Specific:** not "good performance" but "accurate sentiment classification".
- **Measurable:** numbers, or well-defined qualitative scales applied consistently.
- **Achievable:** based on benchmarks, prior experiments or expert knowledge.
- **Relevant:** aligned with the application's purpose and its users.
- **Bad:** "Safe outputs". **Good:** "Less than 0.1% of outputs out of 10,000 trials flagged for toxicity".

> ON THE EXAM
>
> Anthropic puts success criteria at the start: define them, then design evaluations to measure against them.

Dimensions to set criteria on

DIMENSION	THE QUESTION IT ANSWERS
Task fidelity	How well must it do the task, edge cases included?
Consistency	How similar must answers to similar inputs be?
Relevance and coherence	Does it address what was actually asked?
Tone and style	Does it suit the audience?
Privacy preservation	How must it handle personal or sensitive data?
Context utilisation	Does it use the context and history it is given?
Latency and price	How fast, and at what cost per call and per month?

ON THE EXAM

Real criteria are multidimensional: a quality score, a safety rate and a response-time target, together.

Questions that expose hidden requirements

- **Who acts on the output,** and how much do they trust it?
- **What does a wrong answer cost,** and which kind of wrong is worse?
- **When is the peak,** and what happens if the service is down?
- **Whose data is it,** and what do contracts and regulators say about it?
- **What must it never do,** whatever the user asks?
- **Who signs it off,** and what evidence will they want?

> THE TAKEAWAY
>
> Hidden requirements surface late and expensively. Ask these before the first prompt is written.

Solution first, or requirement first?

SOLUTION FIRST

"Let's build an agent"

- Model, framework and agent chosen up front
- Requirements bent to fit the design
- Nothing to test the choice against
- Complexity nobody asked for

REQUIREMENT FIRST

"What must it achieve?"

- Requirements and criteria agreed
- Simplest design that meets them
- Tested on representative data
- Complexity added only when needed

THE TIEBREAKER

Anthropic: find "the simplest solution possible", and only increase complexity when needed.

Copperlet's repair triage, written down

ID	REQUIREMENT	KIND
R1	Every emergency in the test set is classed as an emergency	Functional
R2	Trade correct on at least 90% of the test set	Functional
R3	Emergencies reach the on-call queue within 60 seconds, all day, every day	Infrastructure
R4	Average cost at most USD 0.04 per ticket, about USD 360 a week	Infrastructure
R5	Harwood tickets reach Claude only through Bedrock in a UK or EU region	Infrastructure
R6	Drafts never promise a repair date; an agent approves every draft	Functional

THE SIGNAL

R1 is deliberately absolute: a missed gas leak costs far more than a false alarm.

Success criteria as tests

```python
# evals/test_triage_requirements.py - one test per requirement ID
def test_r1_every_emergency_flagged(results):
    missed = [r for r in results if r.expected == "emergency" and r.got != "emergency"]
    assert not missed, f"R1: {len(missed)} emergencies classed lower"

def test_r2_trade_accuracy(results):
    assert accuracy(results, field="trade") >= 0.90

def test_r4_cost_per_ticket(results):
    assert mean_cost_usd(results) <= 0.04
```

2-4 R1 has no tolerance: one missed emergency fails the build.

6-10 Thresholds come from the agreed requirements, not from the last run.

THE TAKEAWAY

Criteria written as tests run on every prompt or model change, so drift is caught, not discovered.

Traceability: from need to proof

01

Business need

Faster, safer handling of repair requests.

02

Requirement

R1: every emergency is flagged.

03

Design decision

Urgency as a schema enum; emergency rules in the prompt and in code.

04

Evidence

test_r1 on the labelled test set, on every change.

05

In production

Agents' corrections logged and reviewed weekly.

ON THE EXAM

When an auditor or a new teammate asks why, the trail answers in minutes.

> A head of sales asks a developer to "build an AI assistant for the sales team". The team is keen to start by comparing models and agent frameworks.
>
> No users, inputs, outputs or measures of success have been discussed.

What should the developer do first?

A Choose the most capable model available, so that any later requirement will be met

B Agree with stakeholders who will use it, its inputs and outputs, the decisions it supports, and measurable success criteria

C Build a multi-agent prototype quickly, to show the sales team what is possible

D Adapt a published prompt from a similar sales assistant and refine it later

◆ PAUSE AND DECIDE

What should the developer do first?

A Choose the most capable model available, so that any later requirement will be met

Solution first. Capability does not cover latency, cost or data rules.

B Agree with stakeholders who will use it, its inputs and outputs, the decisions it supports, and measurable success criteria

Requirement first: everything else is chosen against this.

C Build a multi-agent prototype quickly, to show the sales team what is possible

Complexity before anyone knows what is needed.

D Adapt a published prompt from a similar sales assistant and refine it later

Someone else's requirements, not this team's.

THE SIGNAL

"No users, inputs, outputs or measures of success" is the signal: requirements come first.

A support team wants Claude to summarise customer calls for account managers. The draft requirement says: "Summaries should be good and fast."

The developer is asked to turn it into a success criterion the team can test.

Which criterion is best?

A. Summaries should be high quality, and responses should feel quick to account managers

B. Use the most capable model with streaming, so that summaries are as good and as fast as possible

C. On three hundred representative calls, nine in ten summaries rated accurate against a written rubric, no personal data beyond the account ID, and most returned within eight seconds

D. Summaries should be as good as those written by the best account manager

◆ PAUSE AND DECIDE

Which criterion is best?

A. Summaries should be high quality, and responses should feel quick to account managers

Still the goal, reworded. Nothing could fail it.

B. Use the most capable model with streaming, so that summaries are as good and as fast as possible

A design choice, not a criterion.

C. On three hundred representative calls, nine in ten summaries rated accurate against a written rubric, no personal data beyond the account ID, and most returned within eight seconds

Specific, measurable, relevant and multidimensional.

D. Summaries should be as good as those written by the best account manager

Not measurable, and possibly not achievable.

THE SIGNAL

A criterion names a data set, a measure, a threshold, and covers more than one dimension.

> A software company has built its assistant on the first-party Claude API. A new enterprise customer's contract requires that its users' prompts are processed only within the European Union.
>
> The contract is signed, and the customer goes live next quarter.

How should the developer treat this?

A As an infrastructure requirement that can decide the platform, such as a cloud provider's EU regional endpoint for this customer

B As a prompt requirement, by telling Claude in the system prompt to process data only in the European Union

C As a setting, by asking the first-party API to keep inference inside the European Union

D As a later concern, to be revisited after the customer has launched

◆ PAUSE AND DECIDE

How should the developer treat this?

A As an infrastructure requirement that can decide the platform, such as a cloud provider's EU regional endpoint for this customer

Residency is infrastructure, and the platform is where it is met.

B As a prompt requirement, by telling Claude in the system prompt to process data only in the European Union

A prompt cannot move a data centre.

C As a setting, by asking the first-party API to keep inference inside the European Union

The first-party geography setting offers global or US only.

D As a later concern, to be revisited after the customer has launched

A signed contractual requirement, ignored.

CURRENCY CHECK

Where data may be processed is an infrastructure requirement. It is met by platform and region, never by a prompt.

> A retailer needs product descriptions for 200,000 catalogue items before a relaunch in ten days. Nobody waits on any single item. Finance has set a fixed budget, and marketing must approve the quality.
>
> A developer is drafting the design.

Which two design decisions follow from these requirements? Choose two.

A Stream every response, so that descriptions appear quickly

B Process asynchronously, for example with the Message Batches API

C Use a multi-agent system, with a specialist agent per category

D Agree a cost ceiling and a quality bar, and check both on a sample first

E Call the API synchronously, one item at a time, to keep it simple

◆ PAUSE AND DECIDE

Which two design decisions follow from these requirements? Choose two.

A Stream every response, so that descriptions appear quickly

Streaming serves a waiting person. Nobody waits.

B Process asynchronously, for example with the Message Batches API

No one waits, budget fixed: Batch halves the cost.

C Use a multi-agent system, with a specialist agent per category

Complexity no requirement asked for.

D Agree a cost ceiling and a quality bar, and check both on a sample first

Budget and approval become criteria, tested before spending.

E Call the API synchronously, one item at a time, to keep it simple

Simple, but slow, full price and hard on rate limits.

THE SIGNAL

"Nobody waits" plus "fixed budget" points to Batch. "Must approve the quality" points to a sample check.

> A contract-review tool flags risky clauses for a legal team, and the demo goes well. Before launch, the developer asks who acts on the output. The answer: lawyers plan to skip reading contracts the tool marks as low risk.
>
> Testing so far has measured overall accuracy only.

What does this answer reveal?

A Nothing that affects the design, because the demo already showed good overall accuracy

B A hidden requirement: missed risks are costly, so criteria and evaluation must focus on catching risky clauses, with human review designed around that

C A cost problem, so the tool should move to a cheaper model before launch

D A trust problem, so the tool should state that its low-risk ratings can be relied on

◆ PAUSE AND DECIDE

What does this answer reveal?

A Nothing that affects the design, because the demo already showed good overall accuracy

Overall accuracy can hide the misses that matter most.

B A hidden requirement: missed risks are costly, so criteria and evaluation must focus on catching risky clauses, with human review designed around that

Who acts on it, and the cost of being wrong, reshape the criteria.

C A cost problem, so the tool should move to a cheaper model before launch

Nothing in the answer is about cost.

D A trust problem, so the tool should state that its low-risk ratings can be relied on

Overclaims, and makes the risk worse.

THE SIGNAL

"Who acts on the output?" turned a demo into a requirement: which kind of wrong is worse.

> Six months after launch, an auditor asks a team why their assistant never gives legal advice, and how they know it holds. The team believes it is covered, but cannot say which design choice or test supports it.
>
> The team wants to answer such questions quickly in future.

What practice should the team adopt?

A. Strengthen the system prompt so that the no-legal-advice rule is stated more forcefully

B. Write a detailed wiki page describing how the assistant works today

C. Trace each requirement to the design decision that meets it and the test or evaluation that proves it, and keep the links current

D. Keep full production logs, so that answers can be reconstructed whenever someone asks

◆ PAUSE AND DECIDE

What practice should the team adopt?

A Strengthen the system prompt so that the no-legal-advice rule is stated more forcefully

Changes behaviour, not evidence. The auditor still gets no answer.

B Write a detailed wiki page describing how the assistant works today

Describes, but does not prove, and goes stale.

C Trace each requirement to the design decision that meets it and the test or evaluation that proves it, and keep the links current

Traceability: the why and the proof, on demand.

D Keep full production logs, so that answers can be reconstructed whenever someone asks

Logs show what happened, not why or whether it is tested.

THE SIGNAL

"Cannot say which design choice or test supports it" is the signal for traceability.

Module 9 in five lines

- **A goal is not a requirement.** A requirement is something you could fail.
- **Functional and infrastructure:** what it must do, and how and where it must run. The second often decides the design.
- **Success criteria** are specific, measurable, achievable and relevant, across several dimensions.
- **Ask the hidden questions:** who acts, what wrong costs, peaks, data, never-dos, sign-off.
- **Requirement first,** simplest design that meets it, criteria as tests, and a trail from need to proof.

COMING UP

Next: designing the application - one call or several, where code works, and sessions.